U.K. ARVINDAN

B.Tech EP + M.Tech EE | EP21B006
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EDUCATION

- IIT Madras, B.Tech Engineering Physics + M.Tech Electrical Engineering (Minor in Computing) **CGPA: 8.49** *2021 – 2027*
- IITM, B.Sc. Programming & Data Science **CGPA: 9.31** *2021 – 2027*
- PSBB Senior Secondary School (CBSE[1]) — Class XII, **99%** *2021*
- PSBB Senior Secondary School (CBSE) — Class X, **97%** *2019*

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR 1601** in JEE Advanced (in about **1,50,000** students) and AIR 4929 in JEE Mains (of 10 lakh students). *2021*
- **Top 1%** of Tamilnadu in the International Olympiad of Chemistry Qualifiers; **1 of 317** selected in India. *2021*
- Awarded the prestigious **KVPY** Fellowship by IISc. *2019*
- Secured **AIR 12** overall & **AIR 1** in Mathematics in NSTSE. *2018*

PROFESSIONAL EXPERIENCE

Stripe (SDE Intern) *May '25 – Jul '25*

- Prototyped an **LLM** pipeline surfacing insights over **merchant accounting data**, explaining trends using related ledger fields.
- Orchestrated a multi-stage **SQL** + LLM chain in **Ruby** using a durable **Temporal** workflow; built a shared Databricks harness for prompt debugging and per-call cost analysis.
- Interviewed in-house accountants to refine project scope toward **user need**, grounding insights in analyses they actually run.

RESEARCH EXPERIENCE

Dual Degree Thesis: Transformers & LLMs for Networking *Aug '25 – Present*

(*Under Prof. Venkatesh R.*)

- Built an **ns-3 802.11** simulation pipeline for pre-association AP selection, replaying every deployment once per candidate AP to label passive scans with ground-truth throughput.
- Designed hierarchical transformers over frames within 110 ms dwells, cutting throughput regret from **5.26 to 0.42 Mbit/s** against signal-strength selection on held-out topologies.
- LoRA-adapted **Llama-3.2-1B** to predict throughput from the same scans, then matched its performance with a bespoke encoder trained from scratch at **0.35M** parameters.

POSITIONS OF RESPONSIBILITY

Team Abhiyaan (SWE → Core Team) *Apr '22 – Aug '23*

Bolt – *Golf cart turned **driverless shuttle**.*

- Engineered GPS routing to custom destinations via OSM[2] API, enabling flawless **500 m autonomous** run at CFI[3] Open House.
- Enabled **real-time** path planning using a voxel grid to process **100,000 3D points/s** from a stereo camera efficiently.
- Led **end-to-end** testing of the full navigation stack employing Gazebo simulations and real-vehicle latency trials.

Vikram – *Autonomous robot for US Army's IGVC[4] at Oakland University.*

- Reduced loop closure error **from 1 m to 5 cm** via novel encoder odometry model for differential-drive robots.
- Designed **master node** "heartbeat" for periodic, real-time fault detection and recovery.
- Placed **1st** in Auto-Nav Design and Cybersecurity challenges at IGVC (1st Indian team to win since 1992).

Shaastra Workshop (Instructor) *Jan '23*

- Delivered "Software Systems for Autonomous Vehicles" workshop on the autonomous stack, path planning, and lateral controllers.

Writing Club (Convenor) *May '23 – May '24*

- Conducted **30+ events** with a Rs.1,00,000+ budget; led Saarang events to 150+ registrations.
- Built a **6-coordinator** team and partnered with peer clubs, co-funding events that drew in their audiences.

PROJECTS

Full Stack Library System# [*Github*]

- Developed a full-stack library system with a **Vue.js** SPA, **Flask + SQLite** backend, **Redis** caching, and **Celery** workers for asynchronous tasks.

Noisy Music Genre Classification# [*Github*] [*Demo*]

- Built **deep-learning models** to classify noisy music mashups using log-mel spectrograms; trained CNN, CNN-BiLSTM & EfficientNet, reaching **93.1% macro-F1**.

Optimal Batting Strategy#

- Implemented a KL-UCB multi-armed bandit algorithm to optimize batting order and playstyle, maximizing runs under wicket and ball constraints.

TECHNICAL SKILLS

Python | C++ | HTML/CSS/JS | React | ROS | SQL | Ruby | Awk | Bash | PyTorch | pandas

RELEVANT COURSEWORK

- Machine Learning Techniques & Practices
- Probability & Information Theory
- Multi-armed Bandits
- Programming Concepts with Java
- Database Management Systems
- Modern Application Development
- System Commands (Linux)

MISCELLANEOUS

- **3rd** of **1,384** teams in Mimamsa at IISER Pune. *2023*
- **Discovered a Main Belt Asteroid** using data from NASA's Pan-STARRS observatory. *(AIASC, 2019)*
- Studied **Japanese** as Elective course. (Proficiency ≈ N4)

Course Project 1. Central Board of Secondary Education 2. Open Street Maps 3. Center For Innovation 4. Intelligent Ground Vehicle Competition